Contents

USGS/MDBC PCR Protocol 18S rRNA V9 (Lane-Medlin)

1 PROTOCOL INFORMATION

1.1 Minimum Information about an Omics Protocol (MIOP)

- MIOP terms are listed in the YAML frontmatter of this page.
- See https://github.com/BeBOP-OBON/miop/blob/main/model/schema/terms.yaml for list and definitions.

1.2 Making eDNA FAIR (FAIRe)

- FAIRe terms are listed in the YAML frontmatter of this page.
- See https://fair-edna.github.io/download.html for the FAIRe checklist and more information.
- See https://fair-edna.github.io/guidelines.html#missing-values for guidelines on missing values that can be used for missing FAIRe or MIOP terms.

1.3 Authors

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1.4 Related Protocols

- This section contains protocols that should be known to users of this protocol.
- Internal Protocols: Derivative or altered protocols, or other protocols in this workflow.
- External Protocols: Protcols from manufacturers or other groups.
- Include the link to each protocol.
- Include the version number (internal) or access date (external) of the protocol when it was accessed.

1.4.1 Internal Protocols

PROTOCOL NAME	LINK	VERSION	RELEASE DATE	INTERNAL/EXTERNAL
AOML 'Omics Protocols	https://github.com/aomlomics/protocols	not applicable	ongoing	

1.4.2 External Protocols

PROTOCOL NAME	LINK	ISSUER / AUTHOR	ACCESS DATE
Not applicable			

1.5 Protocol Revision Record

VERSION	RELEASE DATE	DESCRIPTION OF REVISIONS
1.0.0	2025-12-15	Initial release
1.0.1	2025-12-15	Updated YAML front matter - note that details in main text still reflect AOML parameters and have not been updated
1.0.2	2025-12-17	Updated entire protocol to reflect USGS parameters
1.0.3	2026-01-23	Updated assay name
1.0.4	2026-01-23	Updated primer references and other YAML front matter

1.6 Acronyms and Abbreviations

ACRONYM / ABBREVIATION	DEFINITION
USGS	United States Geological Survey
NOAA	National Oceanographic and Atmospheric Administration
MSU	Mississippi State University
NGI	Northern Gulf Institute
PCR	Polymerase chain reaction
eDNA	environmental DNA
NTC	No template control
EtOH	Ethanol

1.7 Glossary

SPECIALISED TERM	DEFINITION
Extraction Blank	A type of negative control to confirm there is no contamination during DNA extractions. Normally an empty is filter extracted and PCR amplified alongside other samples.
No Template Control	A type of negative control during PCR to confirm there is no contamination during the PCR process. Normally nuclease-free water is run in place of DNA on a PCR.

2 BACKGROUND

2.1 Summary

This protocol describes steps for performing PCR for 18S rRNA V9 marker gene regions using eDNA extracted from Sterivex by USGS. The PCR protocol includes a primary PCR step and a secondary PCR step.

2.2 Method description and rationale

This protocol is used for PCR amplifying the 18S rRNA V9 marker gene regions of environmental DNA. It is highly reproducible and can easily be adapted for any number of samples (i.e. a full 96-well plate or a few samples).

2.3 Spatial coverage and environment(s) of relevance

This protocol can be used to amplify the 18S rRNA marker gene region of any eDNA sample.

3 PERSONNEL REQUIRED

One person with molecular biology experience.

3.1 Safety

There are no hazardous chemicals or materials involved in this protocol. Standard lab safety techniques should still be used such as wearing PPE to avoid skin or eye contact.

3.2 Training requirements

Basic molecular biology training is sufficient for this protocol including sterile technique, pipetting small volumes and programming/running PCR thermal cyclers.

3.3 Time needed to execute the procedure

Protocol takes about 4 hours (240 minutes) including thermal cycler run time.

4 EQUIPMENT

For 96-well Plate:

DESCRIPTION	PRODUCT NAME AND MODEL	MANUFACTURER	QUANTITY	REMARK
Durable equipment				

DESCRIPTION	PRODUCT NAME AND MODEL	MANUFACTURER	QUANTITY	REMARK
100-1000 ul Pipette	Eppendorf Research Plus Adjustable-Volume Pipette	Eppendorf	1	Can be substituted with any accurate pipette
10-100 ul Pipette	Eppendorf Research Plus Adjustable-Volume Pipette	Eppendorf	1	Can be substituted with any accurate pipette
0.1-2.5 ul Pipette	Eppendorf Research Plus Adjustable-Volume Pipette	Eppendorf	1	Can be substituted with any accurate pipette
10-100 ul 8-Channel Pipette	Eppendorf Research Plus 8 Channel Pipette	Eppendorf	1	Can be substituted with any accurate pipette
0.5-10 uL 8-Channel Pipette	Eppendorf Research Plus 8 Channel Pipette	Eppendorf	1	Can be substituted with any accurate pipette
Thermal cycler	Mastercycler Nexus Thermal Cycler	Eppendorf	1	Can be substituted with generic
Consumable equipment				
Gloves	Nitrile Gloves, Exam Grade, Powder-free	ULINE	1	(box) Can be substituted with generic
Kim Wipes	KimWipe Delicate Task Wipers	KimTech	1	(box) Can be substituted with generic
96-well PCR Plate		Armadillo PCR Plate, 96-well, clear, clear wells	ThermoFisher	3
PCR Plate Seal	AlumaSeal II Sealing Foils for PCR and Cold Storage	VWR	2	Can be substituted with generic, can use tightly-fitted strip caps in place of seal
1000µL Filter Tips	OT-2 Filter Tips, 1000µL	Opentrons	1	(box) Can be substituted with generic
200µL Filter Tips	OT-2 Filter Tips, 200µL	Opentrons	2	(boxes) Can be substituted with generic

DESCRIPTION	PRODUCT NAME AND MODEL	MANUFACTURER	QUANTITY	REMARK
10 ul Filter tips	TipOne Pipette Tips, 10 uL	TipOne	2	(boxes) Can be substituted with generic
Kapa HiFi Hotstart ReadyMix	Kapa HiFi Hotstart ReadyMix	Roche	0.6	(mL)
Molecular water	Invitrogen RT-PCR Grade Water	Fisher Scientific	1 vial	
Forward Primer - 1391F	18S 1391F Fluidigm Primer	IDT	105	(ul (1uM)) Primer must be diluted from 100uM stocks to 1uM
Reverse Primer - EukBR	18S EukBR Fluidigm Primer	IDT	105	(ul (1uM)) Primer must be diluted from 100uM stocks to 1uM
2X Kapa HiFi Hotstart ReadyMix	2X Kapa HiFi Hotstart ReadyMix	Roche	2.4	(mL)
Index Primer Mix	Illumina Compatible Index Mix	Illumina	0.48	(mL)
Chemicals				
RNase AWAY	RNase AWAY Surface Decontaminant	ThermoFisher Scientific	1	(bottle) Used to sterilize lab surfaces and equipment
EtOH	Ethanol	Generic Brand	1	(wash bottle) Must be molecular grade ethanol
DI water	Deionized water	Generic	900	(mL)

- Description: E.g., "filter".
- Product Name and Model: Provide the official name of the product.
- Manufacturer: Provide the name of the manufacturer of the product.
- Quantity: Provide quantities necessary for one application of the standard operating procedure (e.g., number of filters).
- Remark: For example, some of the consumable may need to be sterilized, some commercial solution may need to be diluted or shielded from light during the operating procedure.

5 STANDARD OPERATING PROCEDURE

5.1 Protocol

5.1.1 Preparation

1. Dilute primers from 100 uM to 1 uM.
2. Set up PCR under hood by wiping off all surfaces, pipettes, and racks with RNase AWAY and UV sterilizing for 5-10 mins.

3. Map out order of samples on 96-well PCR plate. Make sure to leave a space for a no template control (NTC) and positive control.

5.1.2 PCR

1. Add the following volumes to each well of a 96-well plate:

- 6.25 uL Kapa HiFi Hotstart ReadyMix
- 2.5 ul Fwd primer (1 M) - GTACACACCGCCCGTC
- 2.5 ul Rev primer (1 M) - TGATCCTTCTGCAGGTTCACCTAC

PCR Primer Name	Direction	Sequence (5' -> 3')	Sequence (5' -> 3') with Illumina tail
1391F	forward	GTACACACCGCCCGTC	TCGTCGGCAGCGTCAGATGTGTATAA xxx GTACACACCGCCCGTC
EukBr	reverse	TGATCCTTCTGCAGGTTCACCTAC	GTCTCGTGGGCTCGGAGATGTGTATA xxx TGATCCTTCTGCAGGTTCACCTAC

2. Add 2.0 ul of sample DNA (or molecular water for NTC) to respective wells for a total reaction volume of 13.25 ul per well. Pipette up and down to fully distribute DNA into master mix.
3. Seal plate with PCR plate seal or strip caps.
4. Load plate onto thermal cycler and select program to run the following steps:

PCR step	Temperature	Duration	Repetition
Initial Denaturation	95°C	180s	1x
Denaturation	95°C	30s	30x
Annealing	63°C	30s	30x
Extension	72°C	30s	30x
Final Extension	72°C	5min	1x
Hold	12°C	∞	

5.1.3 PCR 2

Note: Be sure to record which indices are used!

1. Add the following volumes to each well of a 96-well plate:

- 25 uL 2X Kapa HiFi Hotstart ReadyMix
- 5uL Index Primer Mix

2. Add 20 uL of amplified DNA sample to respective wells for a total reaction volume of 50 uL per well. Pipette up and down to fully distribute DNA into master mix.
3. Seal plate with PCR plate seal or strip caps.
4. Load plate onto thermal cycler and select program to run the following steps:

PCR step	Temperature	Duration	Repetition
Initial Denaturation	95°C	180s	1x
Denaturation	95°C	30s	8x
Annealing	55°C	30s	8x
Extension	72°C	30s	8x
Final Extension	72°C	5min	1x
Hold	12°C	∞	

5.2 Basic troubleshooting guide

Low Volume Post-PCR

- If using strip-caps, ensure they are tightly fitting on wells. Any gap in the lid will allow for some volume to evaporate during the PCR process on the thermal cycler. If using PCR plate seals, spin down the plate after taking it off the thermal cycler to ensure all condensation is drawn back into the well.

Contamination

- If there are contamination bands appearing on the gel, run another PCR ensuring full sterilization of work spaces and equipment under the hood and use new vials of Kapa HiFi Hotstart ReadyMix and molecular water. If diluted primers are contaminated, use freshly-made aliquot of primers.

6 REFERENCES

1. Lane, D. (1991). 16S/23S rRNA sequencing. Nucleic Acid Techniques in Bacterial Systematics, 115–175. https://cir.nii.ac.jp/crid/1570009750603612672
2. Medlin, L., Elwood, H. J., Stickel, S., & Sogin, M. L. (1988). The characterization of enzymatically amplified eukaryotic 16S-like rRNA-coding regions. Gene, 71(2), 491–499. https://doi.org/10.1016/0378-1119(88)90066-2
3. Stoeck, T., Bass, D., Nebel, M., Christen, R., Jones, M. D. M., Breiner, H.-W., & Richards, T. A. (2010). Multiple marker parallel tag environmental DNA sequencing reveals a highly complex eukaryotic community in marine anoxic water. Molecular Ecology, 19 Suppl 1, 21–31. https://doi.org/10.1111/j.1365-294X.2009.04480.x

7 APPENDIX A: DATASHEETS

Not applicable.